Econ 1710 $\boxed{\#11}$ 2 April 2021 1 pm Q+A

* Ch 20 intro to options

Everything is DESCRIPTIVE so far. describe product and its features

1. review payoffs - Nika's diamond
2. Call vs. buy stock (intuition)

   ASK questions!

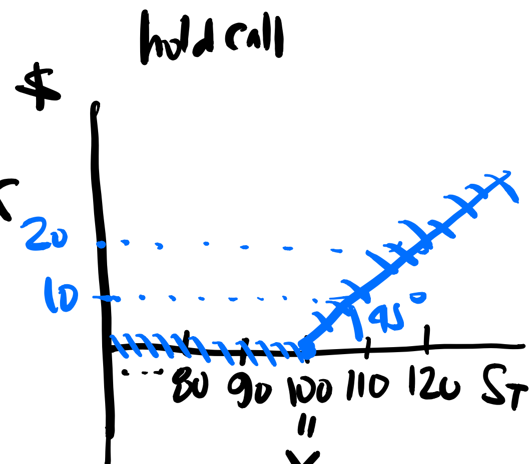
3. option strategies = an algorithm to do
4. put-call parity

$X = 100$

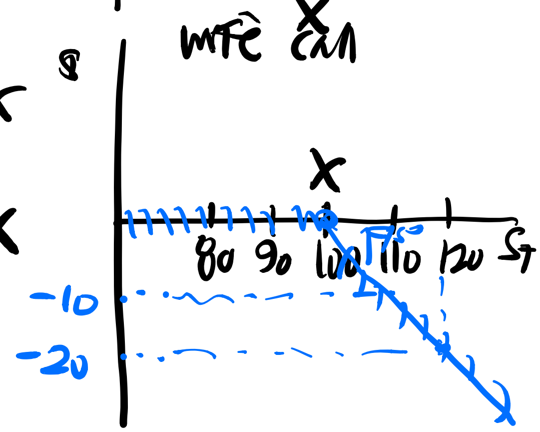
$S_T = 80, 90, 100, 110, 120$

Call is right to Buy

$$\text{Call holder payoff} = \begin{cases} S_T - X & \text{if } S_T > X \\ 0 & \text{if } S_T \le X \end{cases}$$



$$\text{Call writer payoff} = \begin{cases} -(S_T - X) & \text{if } S_T > X \\ 0 & \text{if } S_T \le X \end{cases}$$



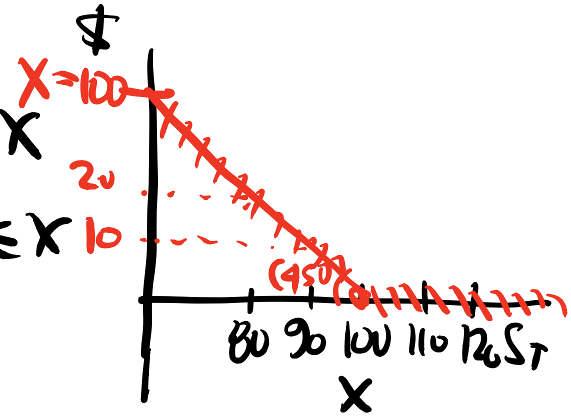
profit = payoff - cost

$X = 100$

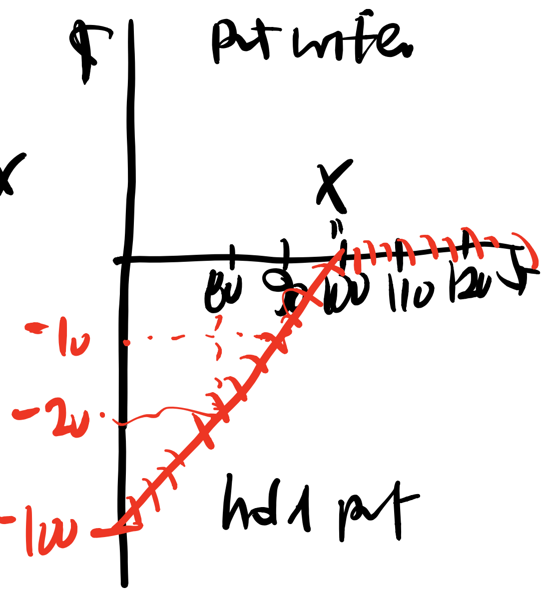
$S_T = 80, 90, 100, 110, 120$

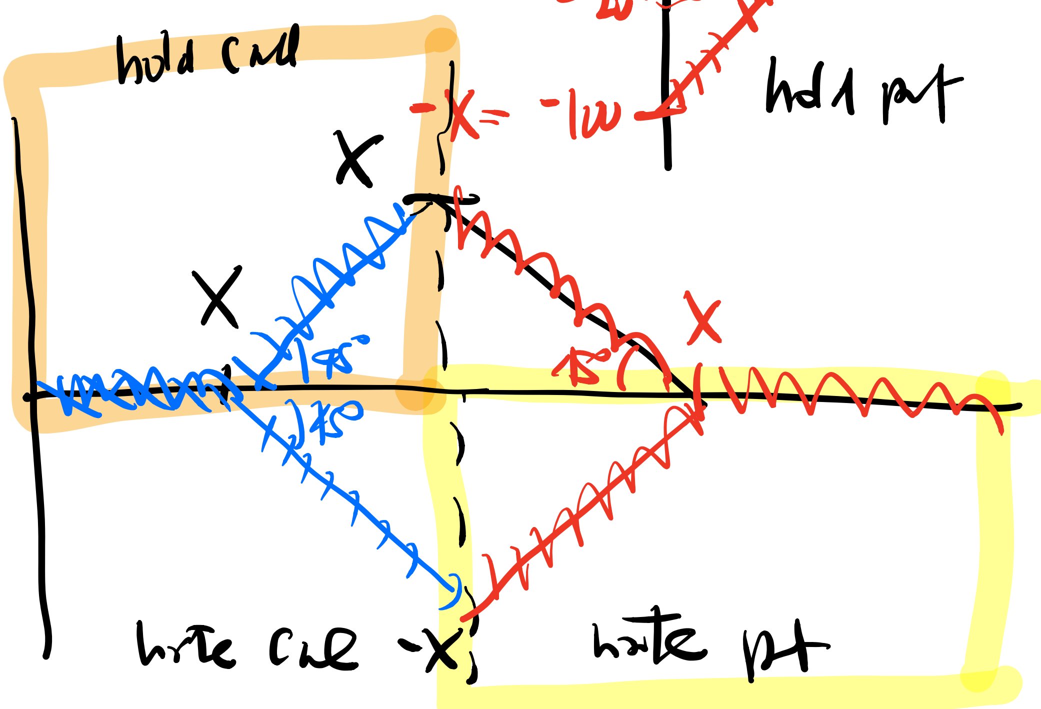
put is the right to SELL

$$\text{put holder's payoff} = \begin{cases} 0 & \text{if } S_T > X \\ X - S_T & \text{if } S_T \le X \end{cases}$$



$$\text{put writer payoff} = \begin{cases} 0 & \text{if } S_T > X \\ -(X - S_T) & \text{if } S_T \le X \end{cases}$$





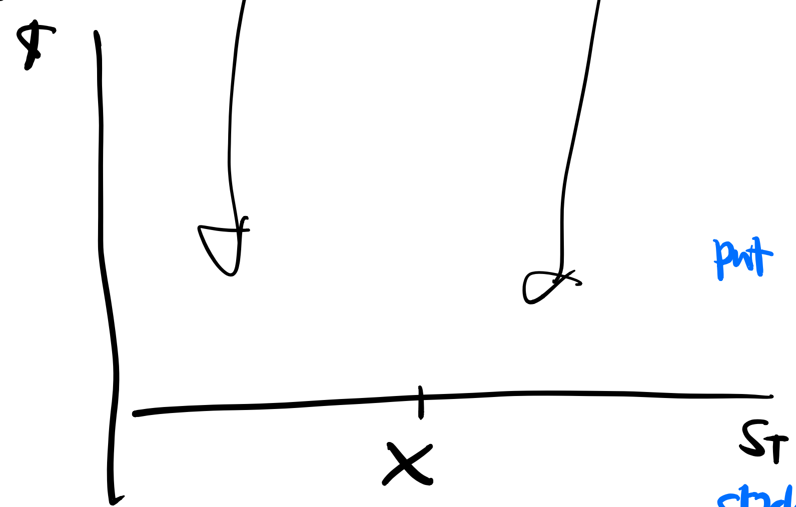
② call v. stock.

③ option strategies.

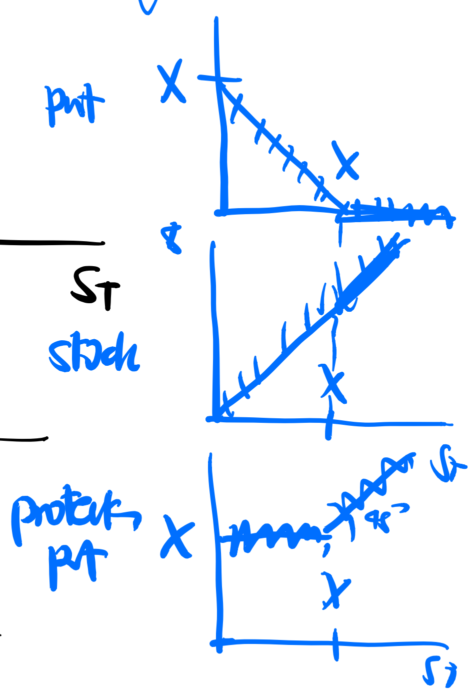
step 1: find payoff

	$S_T \le X$	$S_T > X$
	X_1	X_2
action 1		
action 2		
+		

step 2: graph it!



graphically add.



③ step 3: profit = payoff − cost

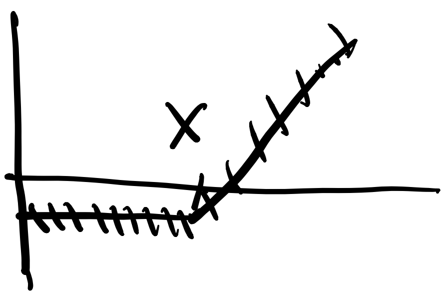
ADD costs of strategy:

action 1 cost

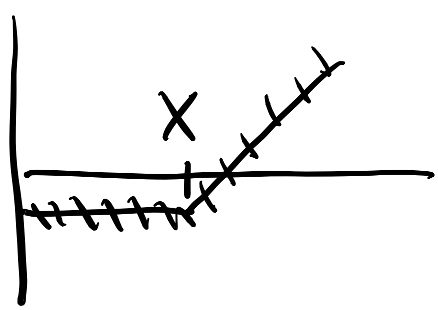
action 2 cost.

"parallel shift" of payoff graph by "cost"

④ Put-Call parity



protective put = 1 share + 1 put



Call plus t-bills = 1 call + lend

same payoff ⟹ same cost

$$C + \underbrace{\frac{X}{(1+r_f)^T}}_{PV(X)} = S_0 + P - PV(div)$$

$$P = C \boxed{+} PV(X) - S_0 \boxed{+} PV(div)$$

$$PV(X) = \frac{X}{(1+r_f)^T}$$

Black-Scholes formula

$$C = [\text{yucky}]$$

- continuous time
- continuous compounding

$$\boxed{PV(X)} = \frac{X}{e^{rT}}$$